Week 3

Classes, Objects and Constructors

Program 7

Write a Java program to create a class Student with data members and display student information using an object.

```
class Student {
   int rollNo;
   String name;
   String course;
   double marks;

   void display() {
      System.out.println("Student Information");
      System.out.println("Roll Number: " + rollNo);
      System.out.println("Name: " + name);
      System.out.println("Course: " + course);
      System.out.println("Marks: " + marks);
   }

   public static void main(String[] args) {
      Student s1 = new Student();

      s1.rollNo = 101;
      s1.name = "Rahul";
      s1.course = "B.Tech CSE";
      s1.marks = 85.5;

      s1.display();
   }
}
```

OUTPUT:
Student Information
Roll Number: 101
Name: Rahul
Course: B.Tech CSE
Marks: 85.5

Program 8

Write a Java program demonstrating default and parameterized constructors.

```
class Student {
   int rollNo;
   String name;

   Student() {
```

```
        rollNo = 0;
        name = "Not Assigned";
    }

    Student(int r, String n) {
        rollNo = r;
        name = n;
    }

    void display() {
        System.out.println("Roll Number: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println();
    }

    public static void main(String[] args) {
        Student s1 = new Student();
        Student s2 = new Student(101, "Rahul");

        System.out.println("Student 1:");
        s1.display();

        System.out.println("Student 2:");
        s2.display();
    }
}
```

OUTPUT:
Student 1:
Roll Number: 0
Name: Not Assigned

Student 2:
Roll Number: 101
Name: Rahul

Program 9
Write a Java program illustrating method overloading to calculate the area of a square, rectangle, and circle.

```
class Area {
    double calculateArea(int side) {
        return side * side;
    }

    double calculateArea(int length, int breadth) {
        return length * breadth;
    }
```

```
    double calculateArea(double radius) {
        return 3.14 * radius * radius;
    }

    public static void main(String[] args) {
        Area a = new Area();

        System.out.println("Area of Square: "
                + a.calculateArea(5));

        System.out.println("Area of Rectangle: "
                + a.calculateArea(10, 4));

        System.out.println("Area of Circle: "
                + a.calculateArea(3.0));
    }
}
```

OUTPUT:
Area of Square: 25.0
Area of Rectangle: 40.0
Area of Circle: 28.26